

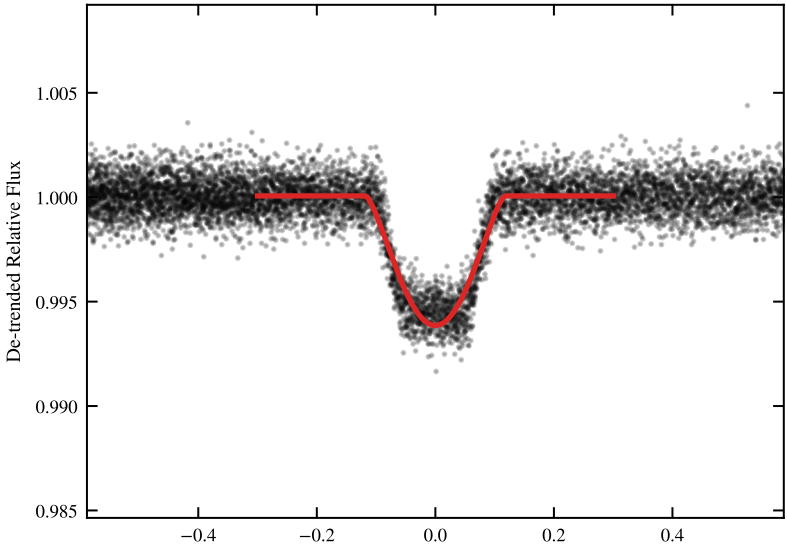
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 461867584 | RA: 149.68855° | Dec: -58.76045° | Sectors: 36, 37

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	4.4208 ± 0.00018
Rp/R★	0.22021 ± 0.0317
Rp (R⊕)	205.22 ± 69.8
b	1.133 ± 0.0357
T14 (h)	5.6431 ± 0.0431
a (AU)	0.11788 ± 0.0355
Teq (K)	6964.1 ± 61.8

Primary Phased Transit



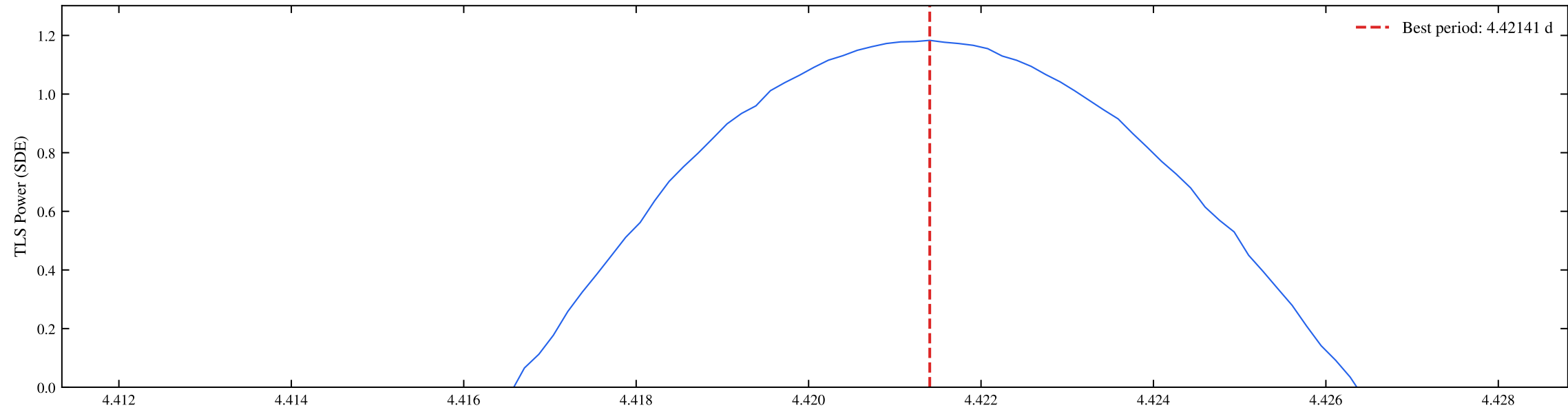
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	4.42141	4.420756 ± 0.00018
P1 Epoch (BTJD)	N/A	2284.2127 ± 0.0011
Max R-hat (R)	N/A	1.0008
Min ESS	N/A	5909
MCMC Divergences	N/A	0

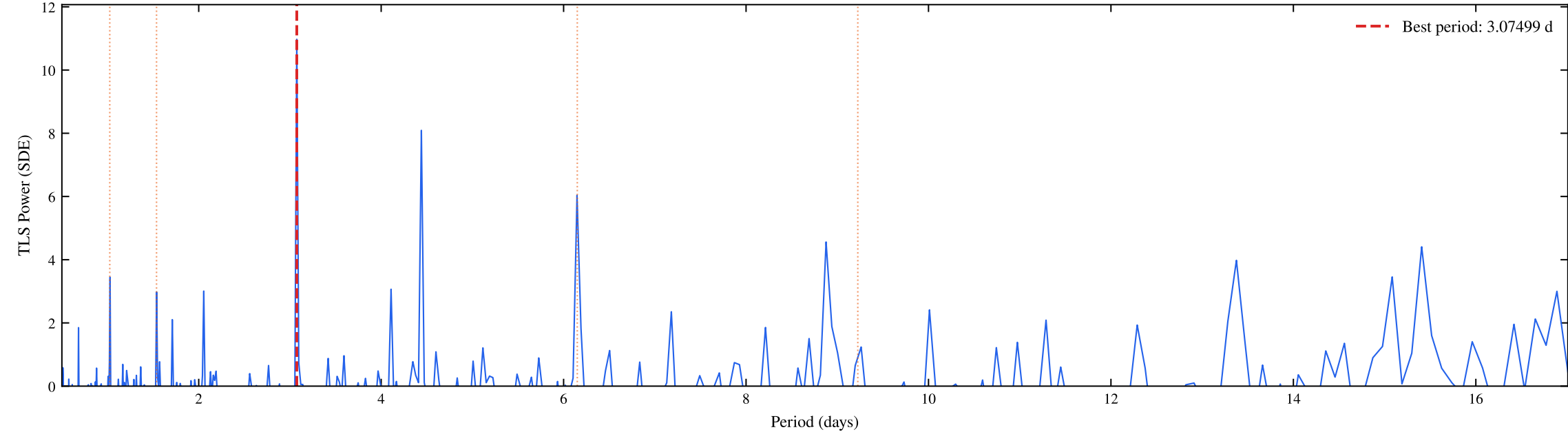
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	8.6 ± 2.58	Derived (Teff scaling)
M★ (M☉)	6.219	Torres 2010
Teff (K)	1.693e+04 ± 134	Gaia DR3
log g (dex)	4.048 ± 0.0508	Gaia DR3
[Fe/H]	0.7717 ± 0.0569	Gaia DR3
ρ★ (g/cm³)	0.01379	Derived

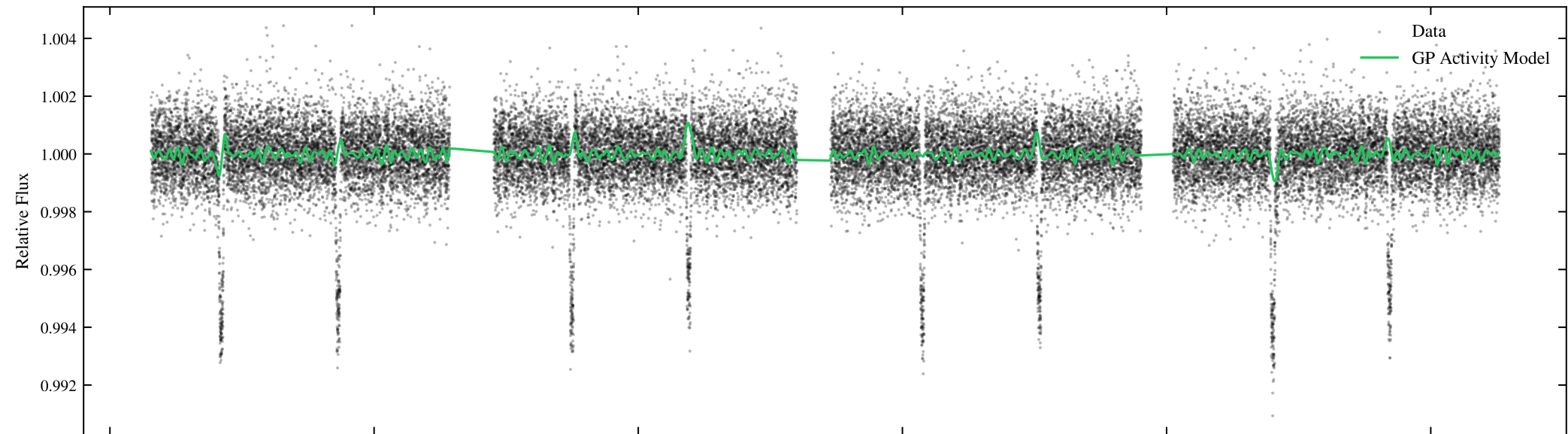
TIC 461867584 | Sectors: 36, 37 | TLS Periodogram (Planet 1 Search) (SDE = 1.18)



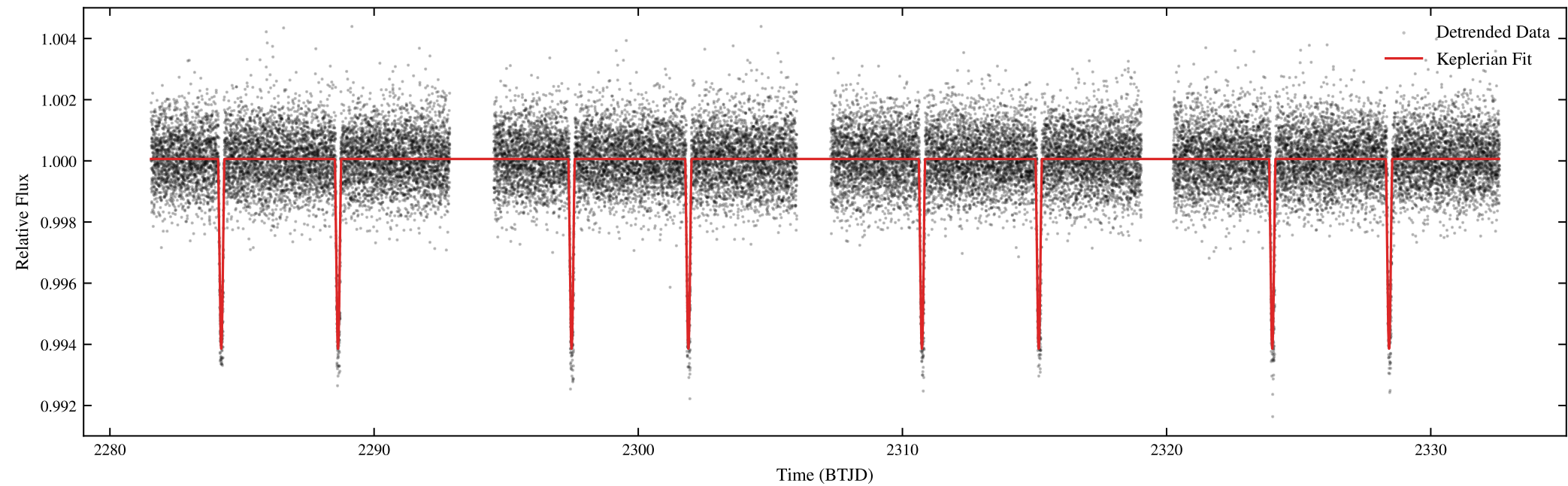
TIC 461867584 | Sectors: 36, 37 | TLS Periodogram (Planet 2 Search) (SDE = 10.97)



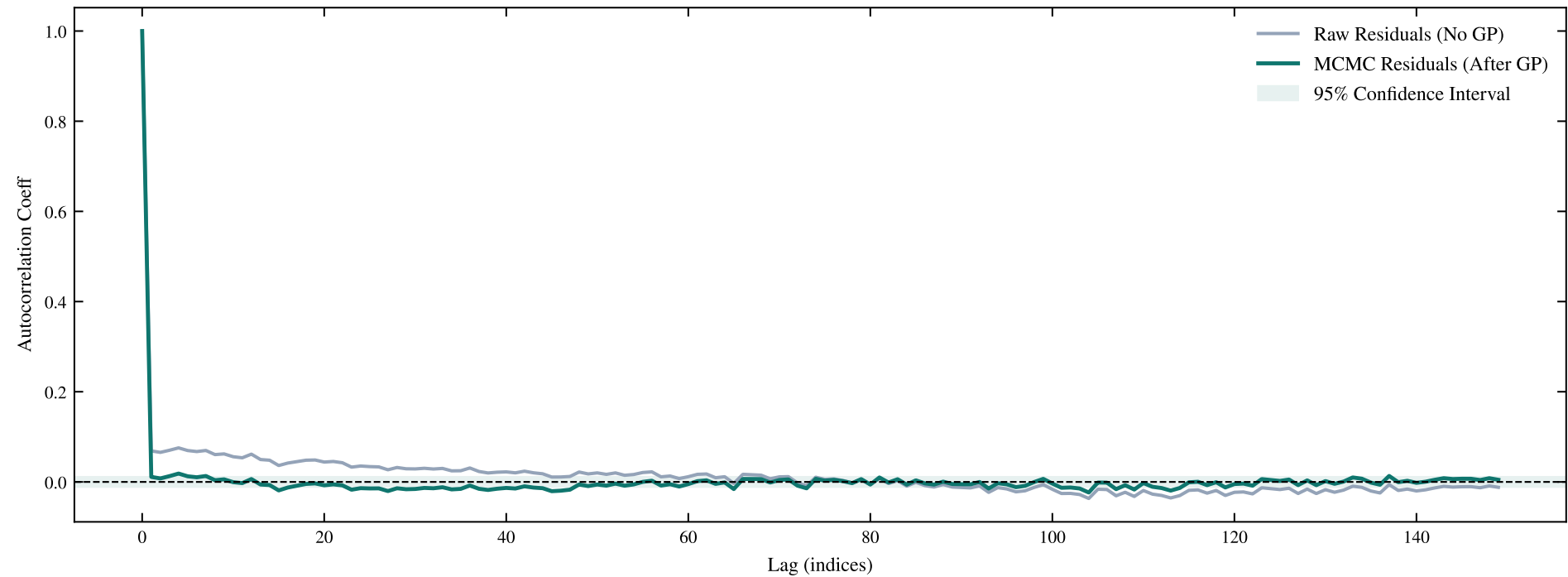
TIC 461867584 | Sectors: 36, 37 | Bayesian GP Detrending (Time Domain)



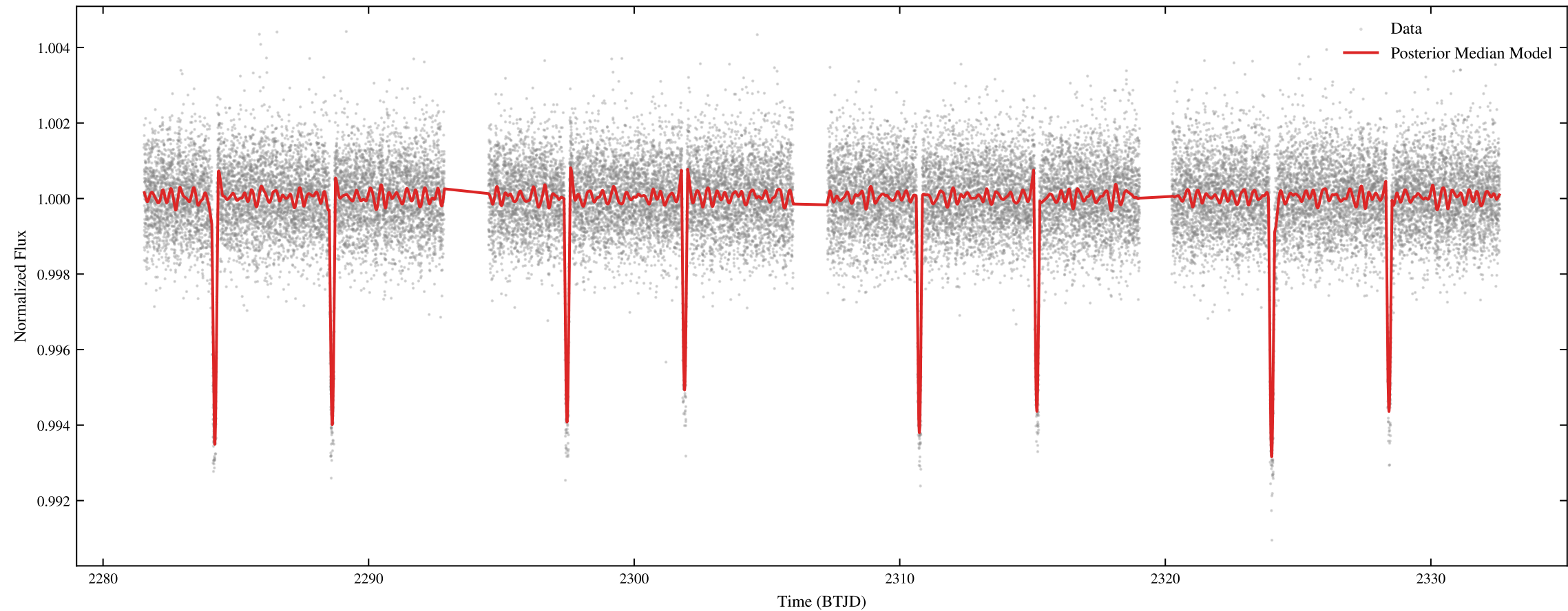
Detrended Light Curve & Best-Fit Transit Model

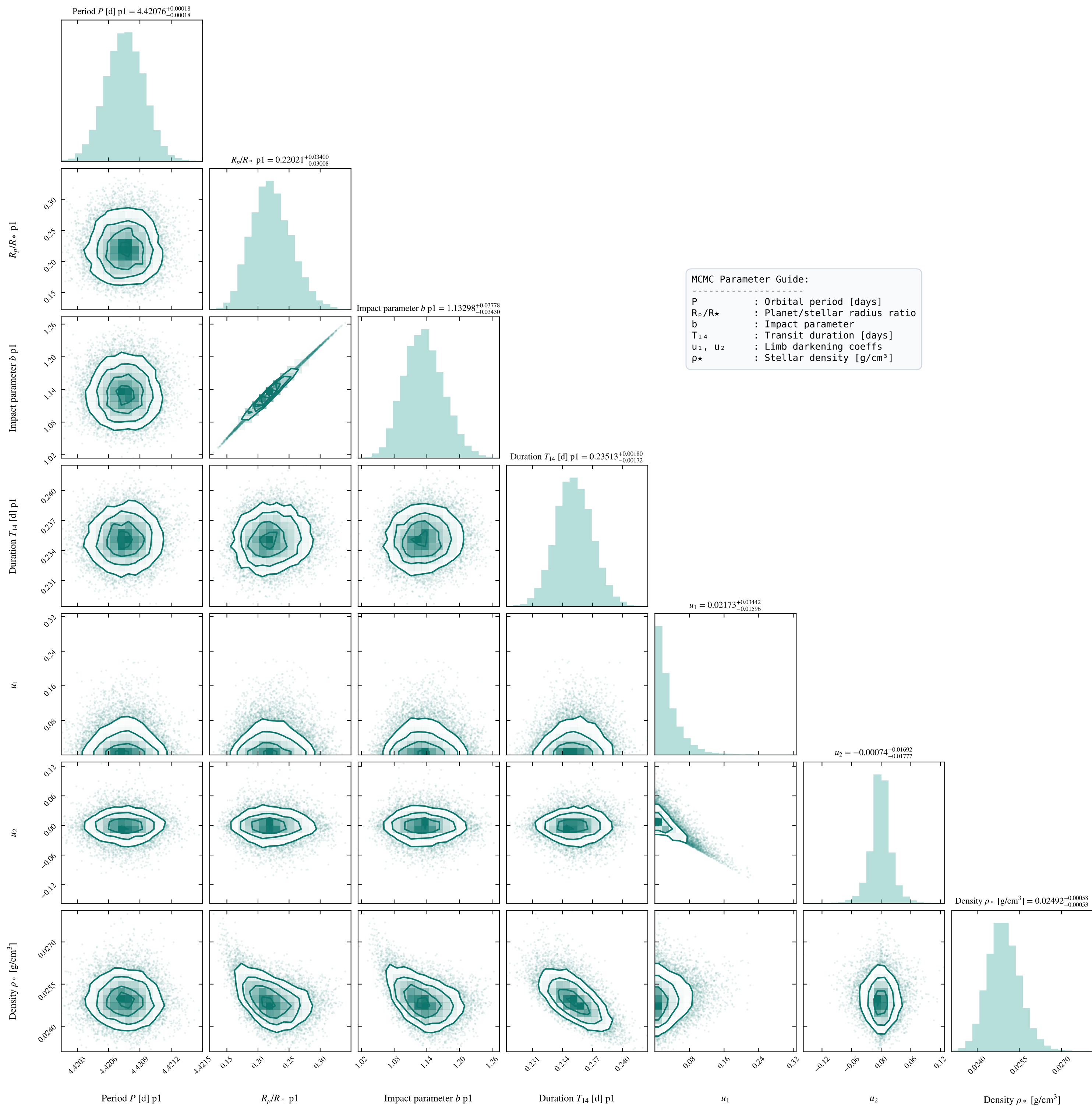


TIC 461867584 | Sectors: 36, 37 | GP Residuals Autocorrelation Function (ACF)

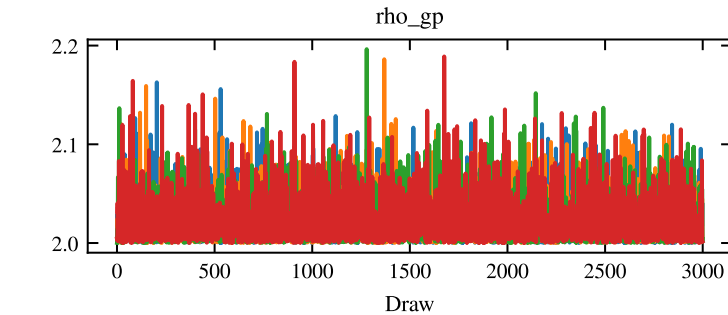
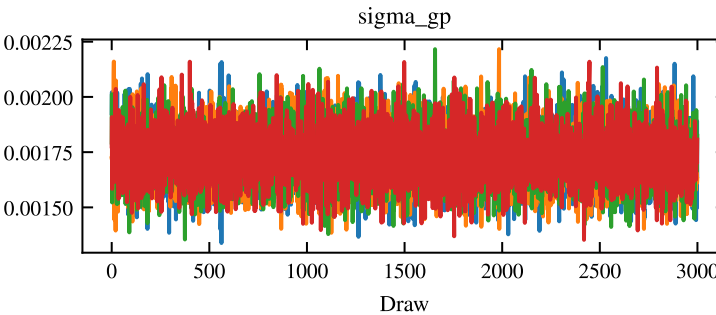
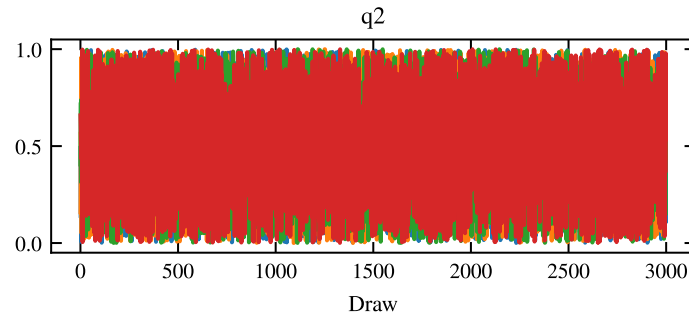
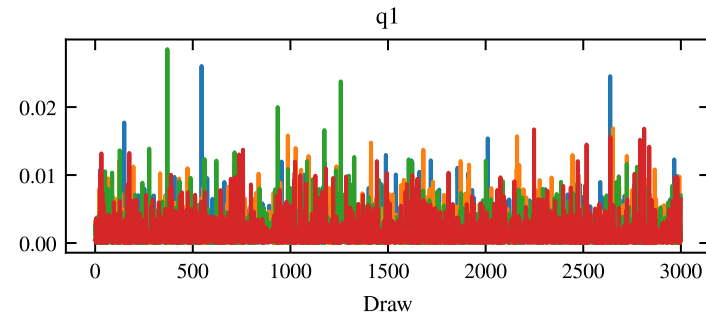
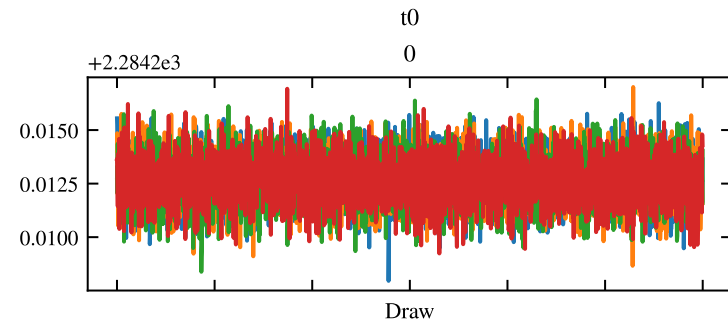
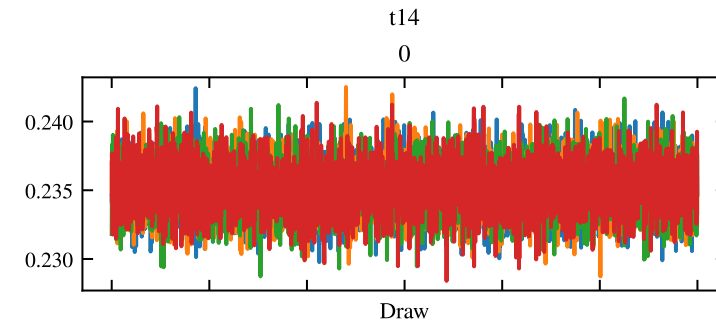
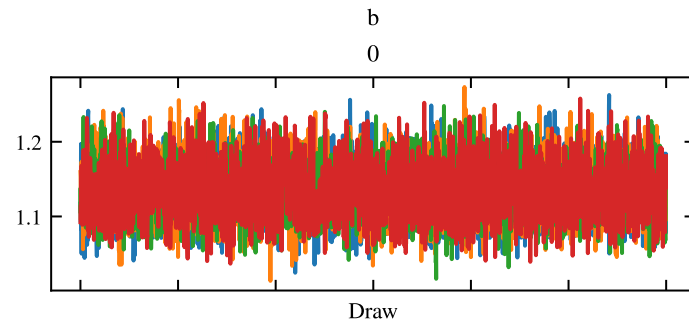
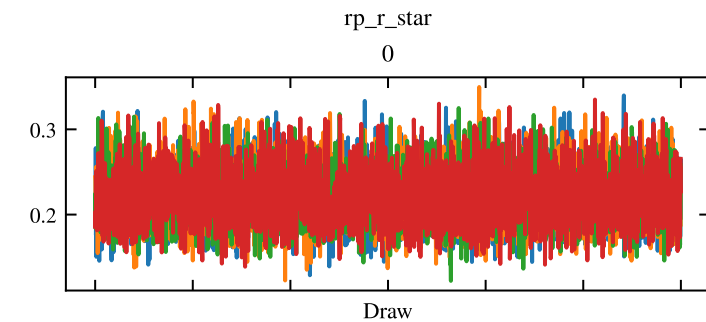


TIC 461867584 | Sectors: 36, 37 | MCMC Posterior Predictive Check

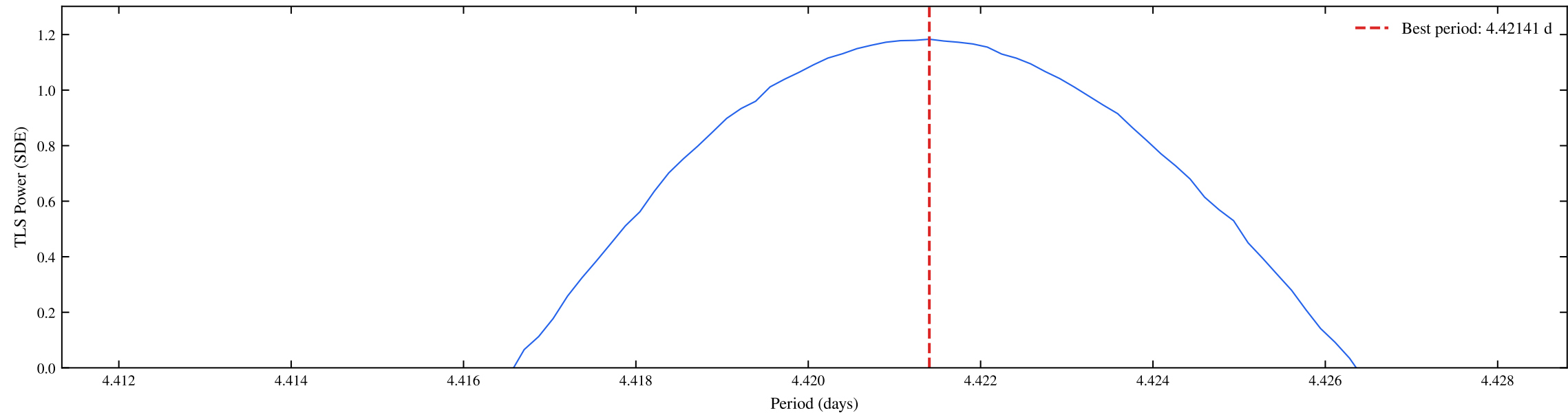




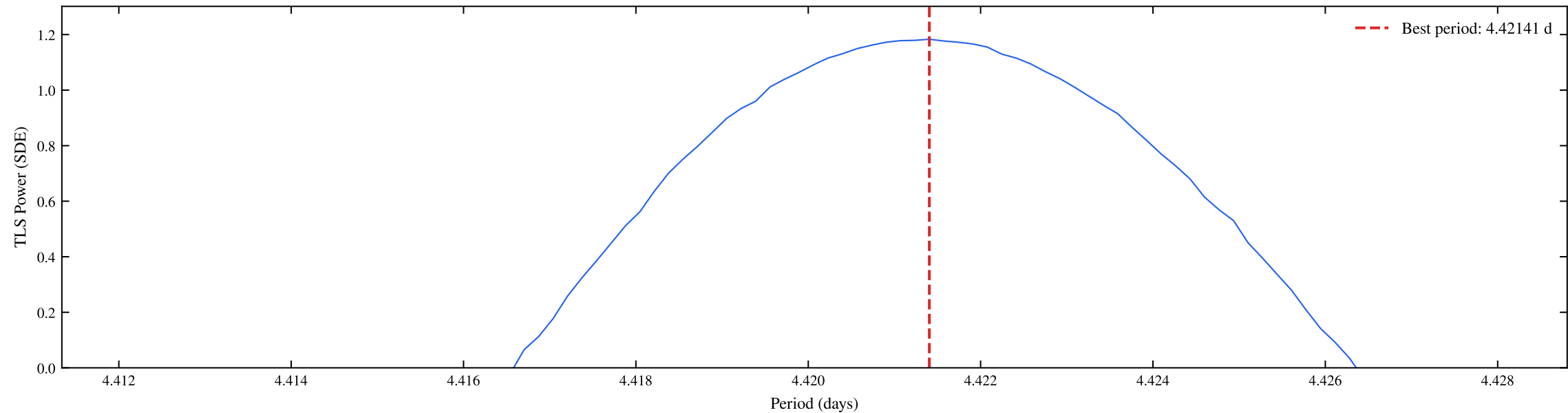
TIC 461867584 | Sectors: 36, 37 | MCMC Trace Plots



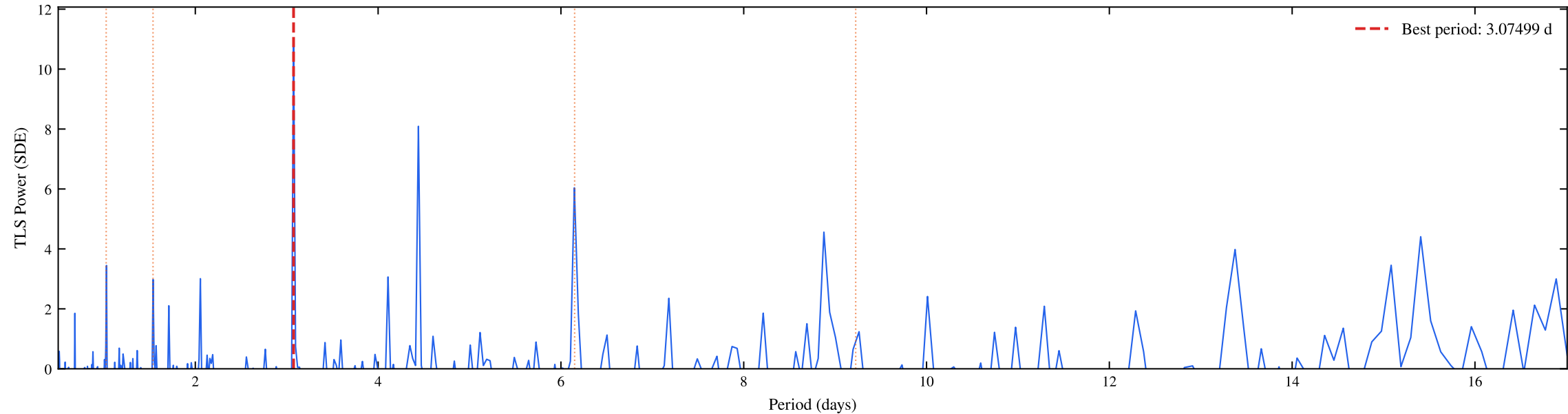
TIC 461867584 | Sectors: 36, 37 | TLS Coarse Periodogram (SDE = 1.18)



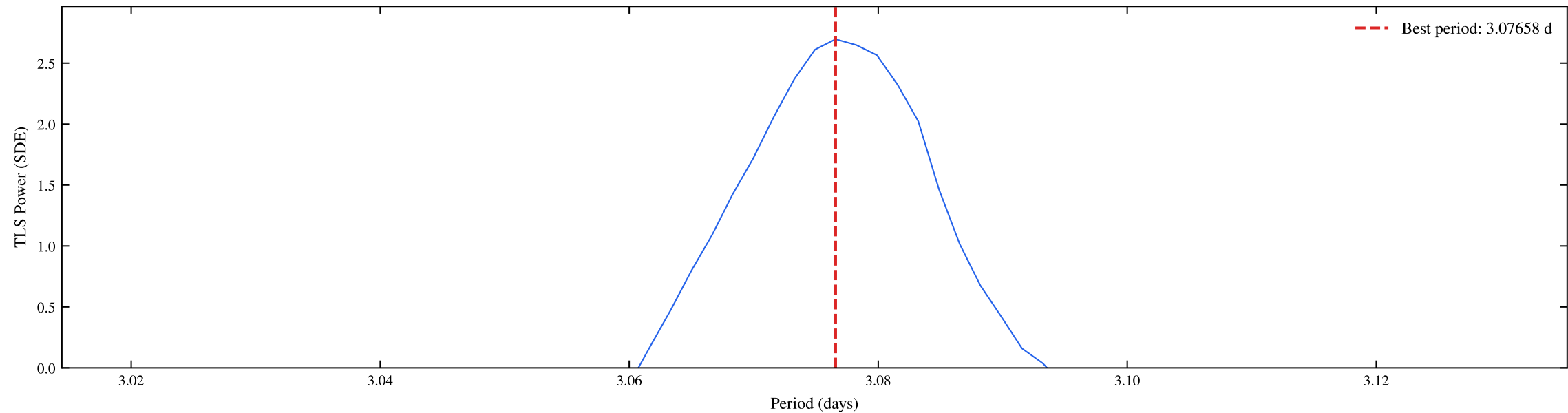
TIC 461867584 | Sectors: 36, 37 | TLS Fine Periodogram (SDE = 1.18)



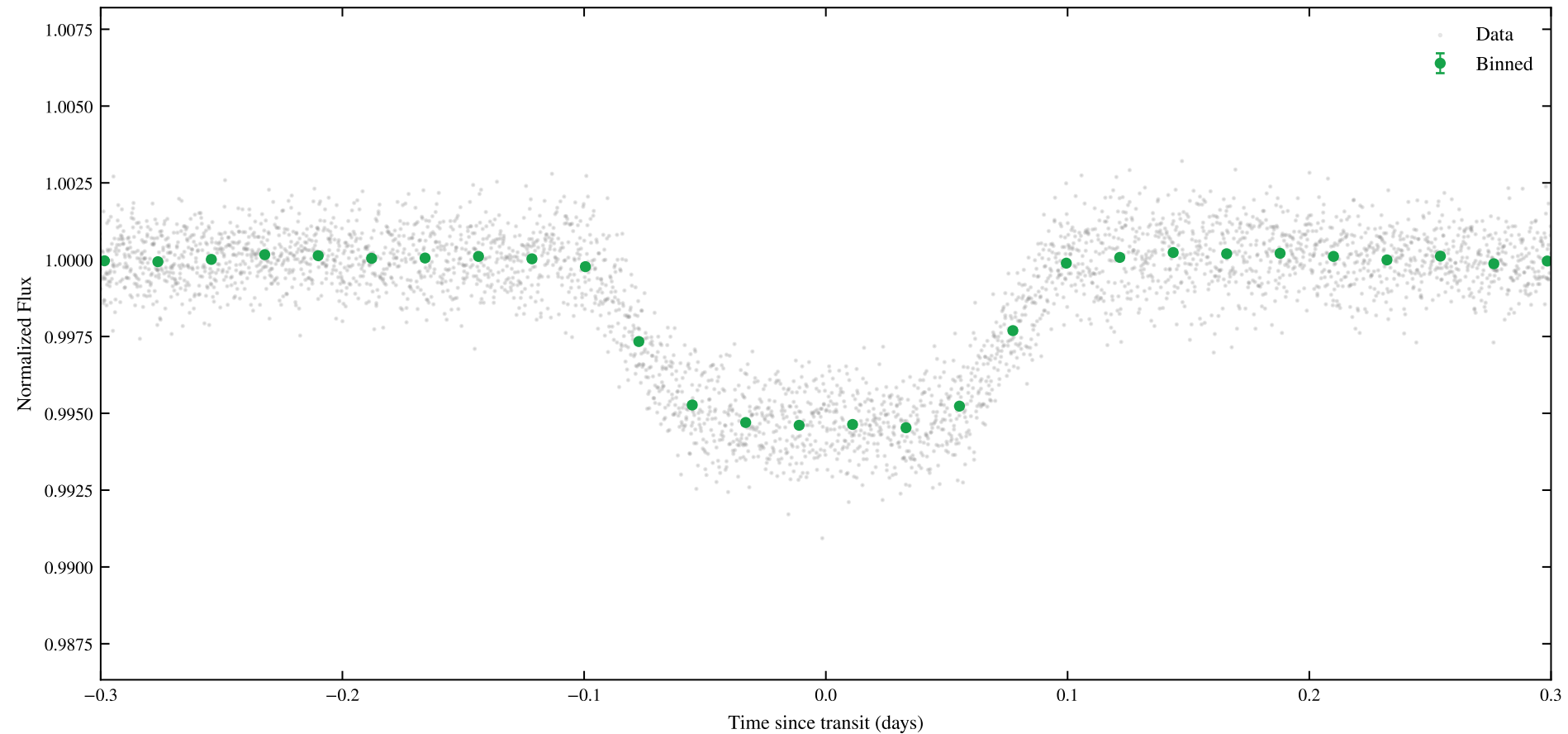
TIC 461867584 | Sectors: 36, 37 | TLS Coarse Periodogram (SDE = 10.97)



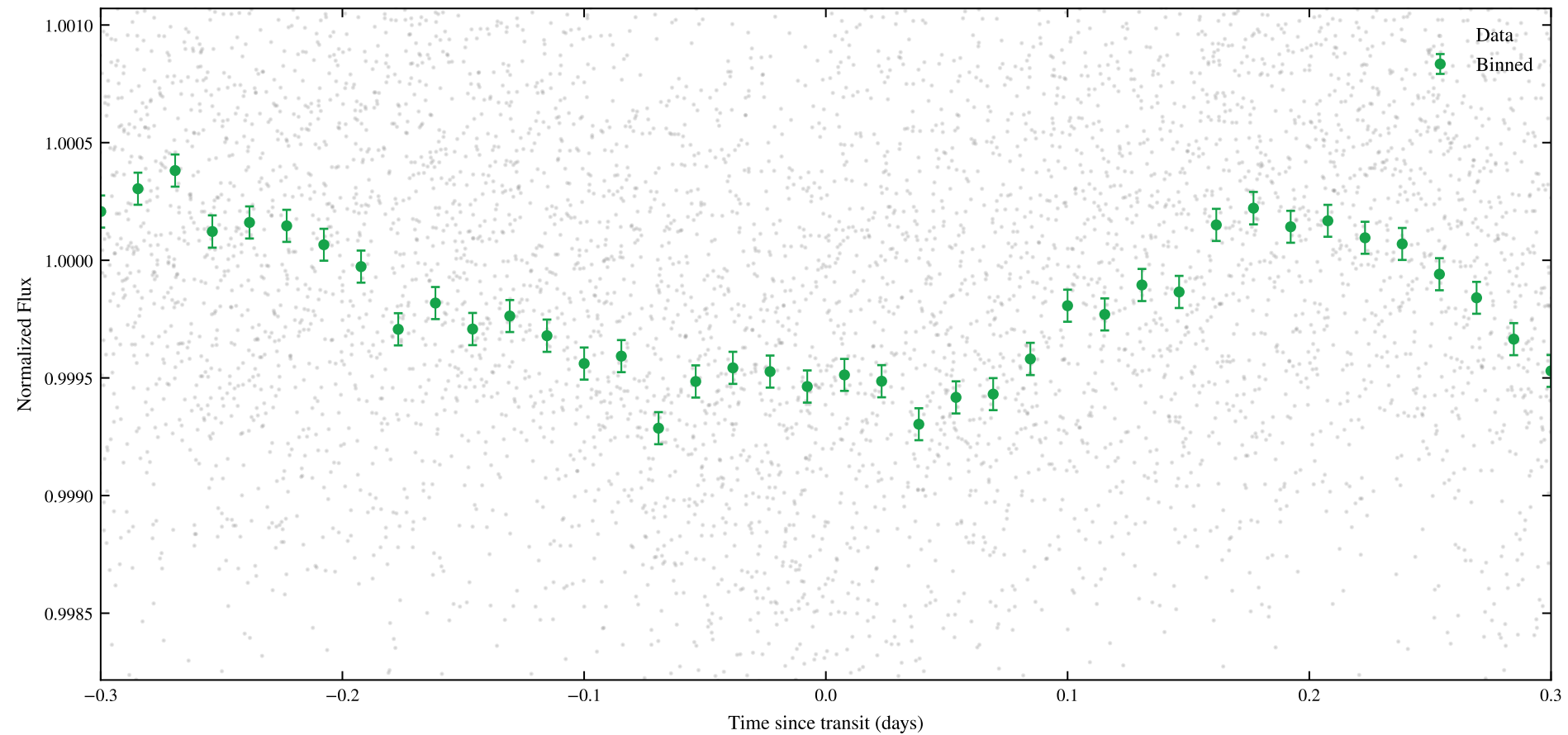
TIC 461867584 | Sectors: 36, 37 | TLS Fine Periodogram (SDE = 2.70)



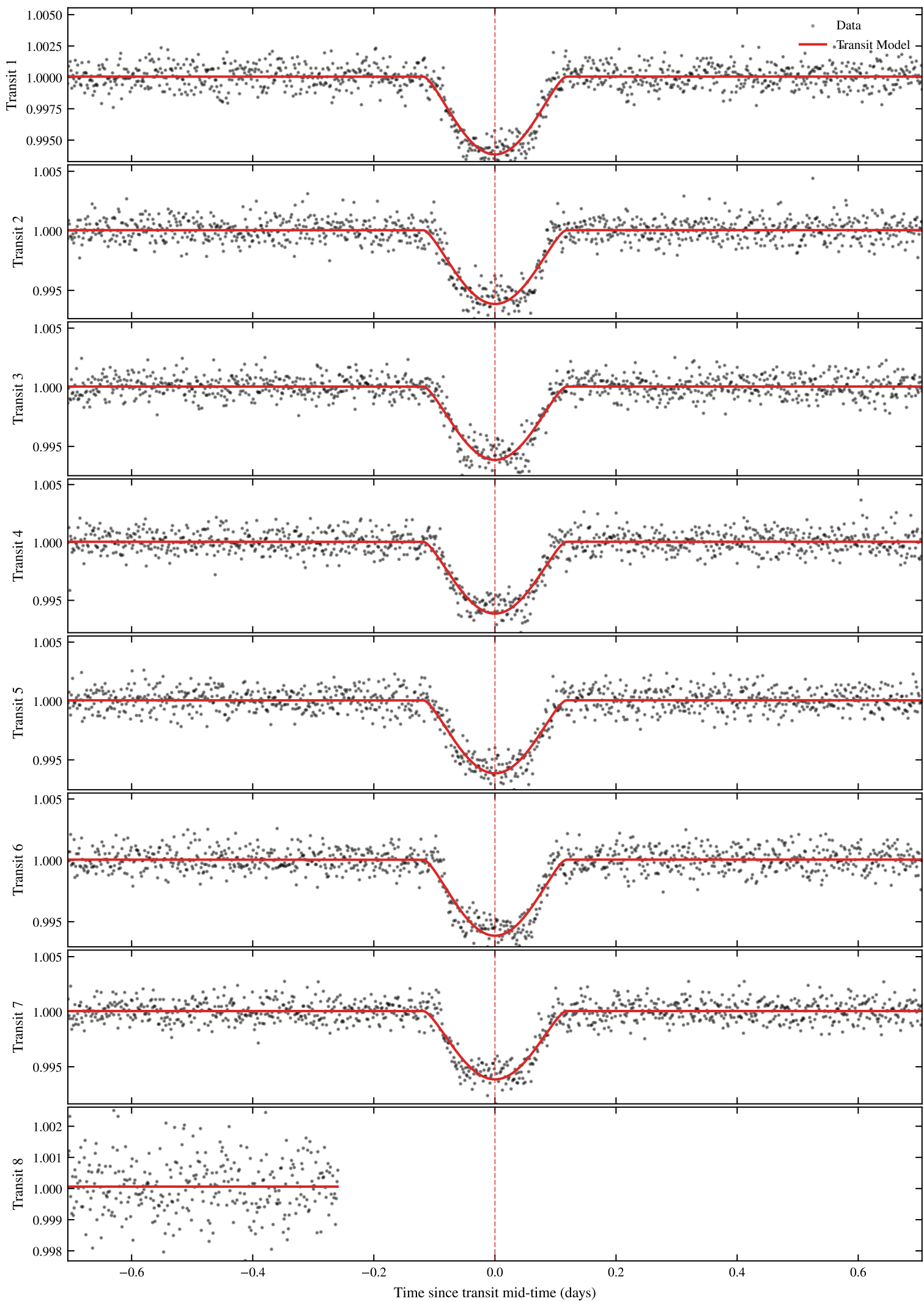
TIC 461867584 | Sectors: 36, 37 | Phase-Folded Transit (P = 4.42141 d)



TIC 461867584 | Sectors: 36, 37 | Phase-Folded Transit (P = 3.07658 d)



TIC 461867584 | Sectors: 36, 37 | Stacked Individual Transits



TIC 461867584 | Sectors: 36, 37 | Phased MCMC Fit - Planet 1

